

FLCR-28 - CAM, Crystal Projectors, And Runtime

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper formalizes CAM/crystal projector mechanics and runtime intake. The operative object is crystal projector. The core result is that CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract. The paper also defines how this result routes forward: this paper becomes the runtime intake lane for agent-generated, standards-window, runtime, notebook-derived, and future crystal work. Its main residue is explicit: private or missing CAM vault data must remain marked as pending rather than assumed.

Keywords

crystal projector; LCR; receipt; claim lane; normal form.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **CAM, Crystal Projectors, And Runtime**. The paper's immediate contribution is: Formalizes CAM/crystal projector architecture, MannyAI task crystals, and agent evidence.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. CAM, Crystal Projectors, And Runtime.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.

- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.
- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines crystal projector as a first-class FLCR object.
- States the local result: CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract.
- Routes downstream use through claim lanes rather than inherited prose: this paper becomes the runtime intake lane for agent-generated, standards-window, runtime, notebook-derived, and future crystal work.
- Preserves the residue boundary: private or missing CAM vault data must remain marked as pending rather than assumed.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- `supplements/CRYSTAL_CAM_PROJECTOR_SUPPLEMENT.md`
- `AGENT_CRYSTAL_WORK_HARVEST.md`
- `PAPER_REWORKS_CRYSTAL_REPORT.md`

Cross-corpus evidence classes:

- `CAM_CLAIM_100_SCORE_AUDIT.md`
- `NSIT_TOOL_CLOSURE_MATRIX.md`
- `NSIT_REASONED_CLOSURE_PASS.md`
- `PAPER_REWORKS_CRYSTAL_PROJECTION.json`
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Crystal projector.** A runtime that loads, projects, and reads addressable crystal/CAM objects through multiple faces.
- **2x2 handshake.** The minimal adapter insertion that preserves address, form, claim lane, and receipt.
- **Receipt boundary.** The exact scope in which the paper's result can be replayed or consumed.
- **Promotion route.** The evidence path required before a stronger downstream claim can use this result.

4. Formal Claims

Claim	Lane	Statement
Theorem 28.1	cam_crystal_reapplication_result	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract
Proposition 28.2	normal_form_result	crystal projector can be imported by later papers only through its declared source, receipt, and lane.
Protocol 28.3	falsifier_or_open_obligation	private or missing CAM vault data must remain marked as pending rather than assumed

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 28.1	CAM route	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 28.2	formal construction	crystal projector can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 28.3	obligation/falsifier	private or missing CAM vault data must remain marked as pending rather than assumed	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-28 defines or uses CAM, Crystal Projectors, And Runtime as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Identify the local object and its assigned sources for FLCR-28.

- 2. Separate what is internally receipt-bound from what is citation-bound, CAM-bound, calibration-bound, or still an obligation.
- 3. State the strongest admissible claim in its current lane.
- 4. Export only the lane-safe result to downstream papers and preserve the residue.

Proof Sketch

The proof strategy for FLCR-28 is a typed construction argument. The paper names crystal projector as the active object, binds it to the routed legacy and orbital sources, and then allows downstream use only through the declared claim lane. This does not erase stronger ambitions; it keeps them addressable as dependencies, calibration tasks, or falsifiers.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is CAM, Crystal Projectors, And Runtime. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by FLCR-01 and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Legacy paper body	Primary formal and source-integration material assigned by the series map.
Internal and pyramid supplements	Cross-paper reasoning windows, deferred back-application slots, and route constraints.
NSIT tool closure matrix	Existing verifier/tool families that can close internal or batch claims.
CAM/crystal and standards-window evidence	Projected memory, source routing, standards reports, and window/node databases.

Standards And Runtime Intake Binding

The runtime instruction suite supplies a runtime doctrine that this paper must own rather than leave as a loose external note. The operative rule is CAM-first: substrate reasoning routes through the CQE CAM crystal before runtime-local CAM state, and runtime-local state is then used for agents, crystals, products, coins, and runtime traces. The paper therefore treats the current runtime MCP surface as an intake adapter in progress: it can expose CAM and Carrier-CEM tools, but full publication closure requires first-routing enforcement in the server path.

The relevant evidence rows are:

Artifact	FLCR-28 use
local evidence artifact	Defines CAM-first routing and the wall/gating discipline this paper imports.
local evidence artifact	Makes runtime errors correction lanes rather than terminal failures.
local evidence artifact	Allows adapter construction by function and receipt rather than by stale names.
local evidence artifact	Current MCP surface; evidence for adapter existence and for the remaining first-routing obligation.
suite_resolution_report.json unresolved row CQE-paper-formal-S29	Routes "The 8 Estimators" into this paper as a CAM/crystal estimator manifest obligation.

Thus Theorem 28.1 is strengthened: a crystal projector is not merely a visual or textual metaphor, but an address-preserving adapter layer whose valid outputs must carry source, face, lane, and receipt fields. It is not yet allowed to claim that every runtime tool already enforces first-routing; that remains a runtime obligation and falsifier until the MCP dispatch path calls the CQE CAM before tool execution.

Appendix A. Recursive Capability Reapplication Review

This review treats the newly admitted capability families as operators. The question is not only what each source says, but what it solves when the source is recursively reapplied through CAM, claim lanes, forge admission, receipt loops, and paper-to-paper routing.

Operator Effects

Operator	Standalone result	Recursive result	Newly resolved state	Remaining boundary
REAPPLY-GLM-CAM-CRYSTAL	The corpus has a working CAM evidence crystal: papers, obligations, verifier findings, download documents, and an integration crystal are addressable nodes with morphisms, ribbons, segmented string windows, and projection queries.	When reapplied, the CAM crystal becomes a paper-brain feedback operator: new findings can be projected into the papers they resonate with, obligations can be reattached to concrete source nodes, and paper-creation feeds can be generated from resonance instead of manual memory.	Local source discovery is no longer only a human search task; it has a content-addressed scan/projection surface.; CAM is no longer only an architectural claim in FLCR-28/40; it has a concrete local crystal that can route evidence and obligations.; The paper suite gains a repeatable way to ask which sources belong in which proof surface.	The local crystal is not yet an exhaustive intake of every future or private source.; Projection resonance is routing evidence, not proof of external physical identity.
REAPPLY-RULE30-ADDRESS-HORIZON	The Rule-30 address horizon supplies a hashed one-megabit center-column reference, a named one-gigabit resource target, and a billion-bit address-space package.	When reapplied, the address horizon becomes a finite prediction and enumeration substrate: CA prediction, process hashing, paper-window addressing, and CAM projection can share a concrete bit-address carrier.	The CA prediction layer now has a finite, content-addressed reference rather than a purely conceptual Rule-30 mention.; Finite horizon claims can be stated strongly without pretending they prove infinite non-periodicity.; Rule-30 data can be used as an addressable carrier for recursive paper and process scans.	The finite reference does not prove infinite non-periodicity.; The one-gigabit Wolfram resource remains a named external acquisition/calibration lane if bit-for-bit reproduction is required.
REAPPLY-PROCESS-CENTROID-AUDIT	The process audit shows that most iterative tools do not explicitly return through a centroid/CAM wrap-back loop.	When reapplied, this becomes an automation falsifier: any future tool or forge claiming closed CAM action must return its result through centroid, receipt, and project-crystal update rather than ending as a one-way script.	A large class of apparent theoretical gaps is reclassified as process hardening: the math is not missing; the wrap-back protocol is missing.; FLCR-39 gains a concrete validator requirement for agent automation.; FLCR-28 gains a negative test for whether a crystal projector is actually closing its own loop.	The audit does not retrofit all 331 missing wrap-back processes.; Each tool still needs an adapter if it is to become a closed CAM action surface.
REAPPLY-GAUSSHAUSE-KERNEL-RING	The kernel ring defines the governance split: kernel as ledger and receipt substrate, ring as admission, input/output gating, validator routing, and obligation propagation.	When reapplied, the kernel ring supplies the 2x2 handshake missing between arbitrary crystal form and actionable CAM lookup: an adapter can admit the object, run the forge, validate the output, and write the child crystal back.	The suite now has an explicit answer to how CAM is actionized into instant lookups and controlled forge calls.; Theory admission, forge admission, and validation authority can share one governance grammar.; The top-layer abstraction problem is narrowed: any form may be addressable if it satisfies the ring's admission and receipt contracts.	The ring governs admission; it does not replace domain validators.; Each target domain still needs its own admissible input vocabulary and promotion criteria.
REAPPLY-GAUSSHAUSE-FORGE-TOPOLOGY	The forge topology describes a typed family of expression surfaces: candidate generation, lattice resolution, CAD/geometry boundary, manufacturing process bridge, SplatForge readout, and simulation/test comparison.	When reapplied, the topology becomes the general expression grammar for the system: every applied paper can say which forge surface is acting, what it reads, what it writes, which validator promotes it, and which residue remains.	Lattice Forge and the forges stop reading as separate tools; they become the way CAM is expressed into domain action.; Materials, proteins, games, plasma, computation, and publication tooling can share a typed read/write/receipt surface.; Physical-realization boundaries become clearer because CAD, process, simulation, printed artifact, and measurement are not collapsed into one result.	The topology does not manufacture external test data.; Every physical claim still requires machine, process, simulation, and metrology receipts where relevant.

Combined Recursive Effects

Combination	Result	Solves
COMBO-CAM-CLOSURE-LEDGER	CAM projection plus closure matrix forms a feedback ledger: evidence can be found, routed, compared to obligations, promoted where bounded, and left as residue where not.	This closes the missing method for moving from discovered crystals to paper-specific obligation updates.
COMBO-KERNEL-FORGE-ACTION	Kernel admission, typed forge action, and centroid wrap-back together define closed CAM action: admit, act, validate, receipt, propagate obligation, and return to the project crystal.	This closes the vague 'forges apply CAM' claim into a concrete operational loop.
COMBO-ADDRESSABLE-PUBLICATION-RUNNER	Rule-30 finite address data, CAM node routing, and paper-runner protocol combine into an addressable publication runner: paper windows and proof artifacts can be enumerated, hashed, routed, and replayed.	This closes the gap between a paper as text and a paper as a state-bound ribbon with executable receipts.

The practical consequence is that several prior gaps are now better classified. Some are closed as routing, admission, finite-address, or executable-publication problems. The residues that remain are sharper: exhaustive source intake, physical calibration, full paper-local runner parity, external measurement, and domain-specific validation.

Appendix B. Broad Capability Intake

This addendum admits non-obvious source material by function rather than filename. Each row states what the source now lets the paper do, while detailed receipt provenance remains in the evidence report instead of the formal claim body.

Capability	Lane	Paper effect	Boundary
CAP-GLM-CAM-CRYSTAL	CAM_crystal_reapplication_result	Promotes CAM from abstract memory to a working evidence crystal: papers, obligations, GLM findings, download documents, and a new integration crystal are content-addressed, ribbon-routed, scanned by segmented strings, and projected back into paper-creation feeds. Evidence facts: 83 content-addressed nodes; 33 formal_paper nodes; 29 open_obligation nodes.	This closes local CAM addressability and routing for the harvested corpus; it does not prove that every future paper source has been exhaustively ingested.
CAP-RULE30-ADDRESS-HORIZON	receipt_bound_internal_result	Adds a concrete address-horizon layer for cellular-automaton evidence: a local one-megabit Rule-30 center-column reference, a named one-gigabit external resource, and a billion-bit address-space package. Evidence facts: Rule-30 local reference: 1000000 bits, 125000 bytes, sha256 4bf71a265b7a3b9847703f274c736c552825a859fd2251d8bfad75b1fe05b3e9; billion-bit address package: 1000000000 bits, 120 chunks, 339 process hashes.	Finite address evidence strengthens prediction and addressability claims; it does not by itself prove infinite non-periodicity or physical identity.
CAP-PROCESS-CENTROID-AUDIT	falsifier_or_open_obligation	Adds negative operational evidence for the process layer: many tools iterate, but very few explicitly wrap their result back through a centroid/CAM return loop. That becomes a design requirement for future agent and forge automation. Evidence facts: 339 processes audited; 8 with centroid/CAM wrap-back; 251 iterative processes; 331 missing wrap-back (97.64 percent).	This is not a failure of the mathematics; it is a process hardening obligation that prevents silent one-way pipelines from being mistaken for closed CAM action.
CAP-GAUSSHAUS-KERNEL-RING	normal_form_result	Adds an explicit admission/governance ring: the kernel owns ledger, hashes, receipts, obligations, and import/export gates, while the ring decides which forge may act, on what inputs, under which evidence commitments. Evidence facts: kernel responsibilities: identity, timeline, source hashing, micro-crystals, receipt chains, obligations, branch/session state, import/export gates; ring responsibilities: admission, input gating, output gating, validator routing, obligation propagation.	This closes policy topology for forge action; domain truth still belongs to the admitted forge, validator, simulation, or test bridge.

Capability	Lane	Paper effect	Boundary
CAP-GAUSSHAUS-FORGE-TOPOLOGY	CAM_crystal_reapplication_result	Adds a typed forge family model: MetaForge proposes candidates, LatticeForge resolves local architecture, CADForge creates inspectable fabrication geometry, the Process Bridge gates manufacturing claims, SplatForge reads the state, and simulation/test bridges compare prediction to observation. Evidence facts: forge surfaces: MetaForge, LatticeForge, CADForge, Process Bridge, SplatForge/GaussHaus Field, simulation/test bridges; typed L/C/R surface: data in/out, control plane, outward receipt surface.	This closes expression architecture for applied materials and forge papers; each physical realization still requires machine, process, simulation, and metrology receipts.

The resulting claim posture is broader but still auditable: a paper can import a capability when the evidence lane is satisfied, and any remaining gap is named as an adapter, calibration, rerun, or falsifier obligation.

Appendix C. Implementation Intake

This addendum binds implementation work that was not fully represented in the earlier paper routing. The rows below are not pasted source text; they are evidence-lane imports that change what the paper can claim now.

Implementation source	Lane	Claim effect	Boundary
IMPL-LCR-TYPED-KERNEL	normal_form_result	Promotes L/C/R from notation to operational lane contract: LAdapter, CKernel, RChannel, strict policy gates, and lane classification.	This closes operational admission and policy routing; it does not by itself prove external physical identity.
IMPL-SPLATFORGE-FIELD-RUNTIME	receipt_bound_internal_result	Adds an implemented crystal-to-spatial-field runtime: deterministic scene/splat hashes, view/semantic operation split, child-crystal fork for semantic edits, and receipt lineage. Verification: Targeted pytest passed: 21 SplatForge field/spatial-adapter tests, including deterministic compile, view/semantic split, child-crystal fork, receipt lineage, render determinism, readonly gating, and promoted-crystal invariance.	Closes desktop data-model behavior; renderer, projector calibration, and AR remain later engineering layers.
IMPL-RENDER-PARITY-GS08	external_calibration_result	Adds CPU/GPU parity evidence for Gaussian/splat rendering: 7 configurations, worst max channel delta 2, all within tolerance <=4. Evidence facts: configs_run=7; worst_max_channel_delta=2; all_within_tolerance=True.	Closes renderer parity for tested configurations; does not validate semantic correctness of source crystals.

The immediate paper effect is stronger claim posture where these implementation rows satisfy the relevant lane. Remaining caution is reserved for specific claims that still lack an adapter, comparator, calibration target, or rerun receipt.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	receipt_bound_internal_result	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.510159.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
bridge/runtime intake	normal_form_result	The bridge action is closed as a correspondence and intake surface with explicit comparator boundaries.	The family closure is a structural lane; measured physics still requires destination-specific calibration.

Row	Lane	Resolved state	Remaining boundary
decade-3 tower	receipt_bound_internal_result	The decade tower is resolved with avg TarPit mass=0.509294 and mass sum=40.743500.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-08 cross-lift comparison	cam_crystal_reapplication_result	Family tower binds FLCR-08, FLCR-18, FLCR-28, FLCR-38; avg TarPit mass=0.510781; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
CAM/crystal projector	cam_crystal_reapplication_result	The crystal projector is closed as an address-preserving multi-face projection protocol.	Private vault contents remain unavailable until attached.

Claim Posture After This Pass

FLCR-28 should now be read as a resolved-state contribution for CAM, Crystal Projectors, And Runtime at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR-11 through FLCR-18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.

- FLCR-28 through FLCR-30 for CAM/crystal, corpus ribbon, and universal normal-form intake.

- FLCR-31 through FLCR-40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- private or missing CAM vault data must remain marked as pending rather than assumed
- External calibration claims require units, datasets, citations, and reproducible data binding.
- A later translation paper may strengthen this result only by adding the missing lane evidence.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): orbital publication overlay.

Original slot	Family	Lift	Role	Proof form
orbital	publication overlay	n/a	no legacy original slot assigned yet	intake and routing proof

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: mature LCR-native corpus and orbital evidence intake.

It must produce: CAM, Crystal Projectors, And Runtime as publication overlay or synthesis route.

Semantic transition: route external, comparative, or synthesis material around the original proof ribbon without overwriting original slot obligations.

Accepted formalism targets to bind in the journal rewrite:

- source routing and dependency graphs
- citation-bound comparison
- normal-form correspondence tables
- falsifier and calibration boundaries

Slot	Contract
orbital	publication overlay; must declare sources, dependencies, and calibration/falsifier boundaries

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work as orbital evidence: route all imports through dependency, citation, normal-form, calibration, and falsifier lanes.

Primary Evidence Inputs

Source	Placement reason
supplements/CRYSTAL_CAM_PROJECTOR_SUPPLEMENT.md	primary assigned source for the paper's formal spine
AGENT_CRYSTAL_WORK_HARVEST.md	primary assigned source for the paper's formal spine
PAPER_REWORKS_CRYSTAL_REPORT.md	primary assigned source for the paper's formal spine

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/CRYSTAL_CAM_PROJECTOR_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/METAFORGE_MATERIALS_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/SPECTRE_TILING_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes

Source	Placement reason
supplements/APPLIED_FORGES_WORKBOOK.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

No direct 00 - 32 CAM score row is assigned; use this paper as an orbital or translation intake.

Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Promote now	Bind next	Residue
none directly assigned	Source routing and claim lanes identify paper-local binding actions.	Verifier catalogs and CAM/crystal routes provide object-name evidence candidates.	Unbound claims remain in falsifier/open-obligation lanes.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Search By Object Name	No direct reasoned-closure queue was assigned from the first queue map.	Search source routing, verifier catalog, CAM/crystal routes, and paper-local object names.	Create a specific binding queue when an object-name match finds a validator, receipt, theorem, or calibration route.	Until then, claims remain in their existing lanes.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
No routed archive card attached yet	0	source-backed contribution	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	0 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR - 28
Legacy source slot(s)	orbital
Ribbon role	routes external, standard-model, or synthesis material around the original proof ribbon
Proof form	source intake, correspondence table, dependency statement, and falsifier boundary
Received state	mature LCR-native corpus and orbital evidence intake
Produced state	CAM, Crystal Projectors, And Runtime as publication overlay or synthesis route

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 28.1	cam_crystal_reapplication_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract
Proposition 28.2	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	crystal projector can be imported by later papers only through its declared source, receipt, and lane.
Protocol 28.3	falsifier_or_open_obligation	Receipt, source card, validator, citation, or CAM route named in the paper manifest	private or missing CAM vault data must remain marked as pending rather than assumed

Figure Plan

Figure	Purpose	Caption
FLCR-28-F1	Publication overlay dependency map	Schematic showing how FLCR-28 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-28-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-28-F3	Same-family lift context	Diagram placing this paper in the original 00-79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-28-F1: Publication overlay dependency map

```

received state
-> Publication overlay dependency map
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

```

Table Plan

Table	Purpose
FLCR-28-T1	Dependency and translation table
FLCR-28-T2	Claim-lane/evidence-boundary table
FLCR-28-T3	Archive evidence card and source-backed upgrade table
FLCR-28-T4	Workbook replay and falsifier table

Worked Example

Route one standard-model or synthesis claim back to its LCR-native source papers. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
dependency	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
publication overlay	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.

Term	Paper-local meaning
synthesis lane	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.
translation claim	Defined by this paper as part of the publication_overlay proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-28-A: Evidence Package

The appendix records:

1. Legacy source excerpts or source-card references used in the final claim ledger.
2. Receipt and verifier paths, including pass/fail/partial status.
3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
4. Standards-window and NSIT evidence used for conformance or routing.
5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-28 (CAM, Crystal Projectors, And Runtime) occupies the publication_overlay role at lift depth n/a. The paper receives mature LCR-native corpus and orbital evidence intake. It produces CAM, Crystal Projectors, And Runtime as publication overlay or synthesis route. The state transition is: route external, comparative, or synthesis material around the original proof ribbon without overwriting original slot obligations.

The paper-local proof form is:

source intake, correspondence table, dependency statement, and falsifier boundary

The integration result is a source-intake and synthesis surface with dependency, citation, calibration, and falsifier boundaries. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
source routing and dependency graphs	Formal carrier for the paper-local result.
citation-bound comparison	Formal carrier for the paper-local result.
normal-form correspondence tables	Formal carrier for the paper-local result.
falsifier and calibration boundaries	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 28.1	cam_crystal_reapplication_result	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract
Proposition 28.2	normal_form_result	crystal projector can be imported by later papers only through its declared source, receipt, and lane.
Protocol 28.3	falsifier_or_open_obligation	private or missing CAM vault data must remain marked as pending rather than assumed

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	supplements/CRYSTAL_CAM_PROJECTOR_SUPPLEMENT.md, AGENT_CRYSTAL_WORK_HARVEST.md, PAPER_REWORKS_CRYSTAL_REPORT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-28-F1: Publication overlay dependency map	Visualizes the proof object, routing, or boundary.
FLCR-28-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-28-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-28-T1: Dependency and translation table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-28-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-28-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-28-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.